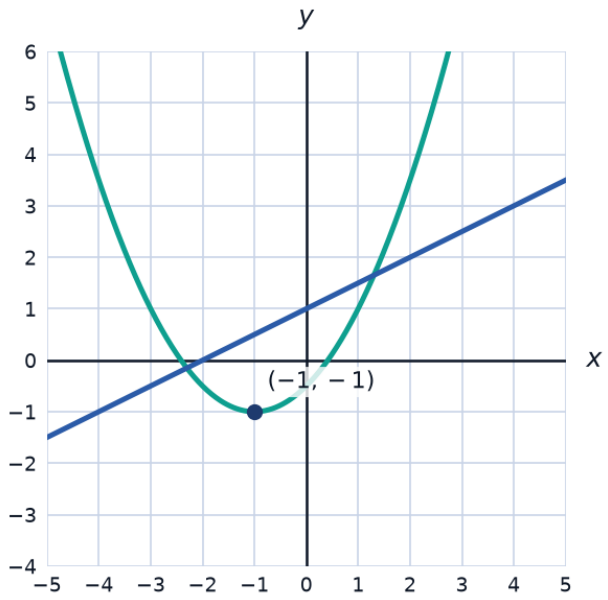
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

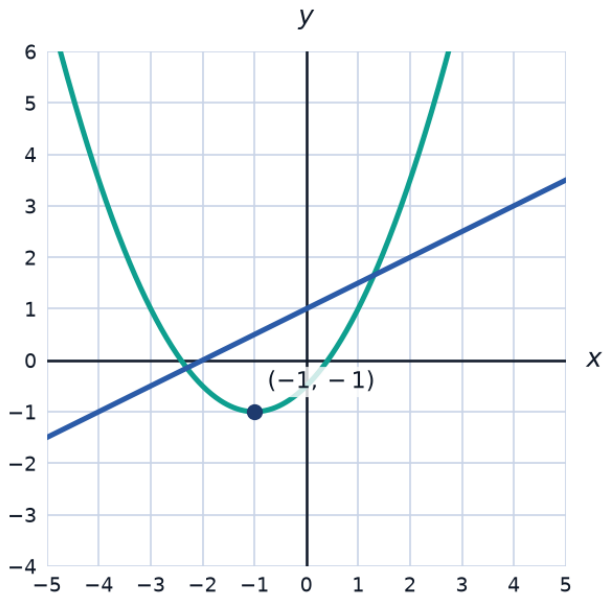
1 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the sum of the infinite geometric series $20 + \frac{20}{3} + \frac{20}{9} + \cdots$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Identify the geometric shape in the complex plane described by the equation $|z - 4| + |z + 4| = 10$, and calculate the length of its minor axis.



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3 [Foundational] • Target: Arithmetic sequences

For the arithmetic sequence with $a_1 = 6$ and common difference $d = 4$, find a_{12}.

4 [Foundational] • Target: Arithmetic sequences

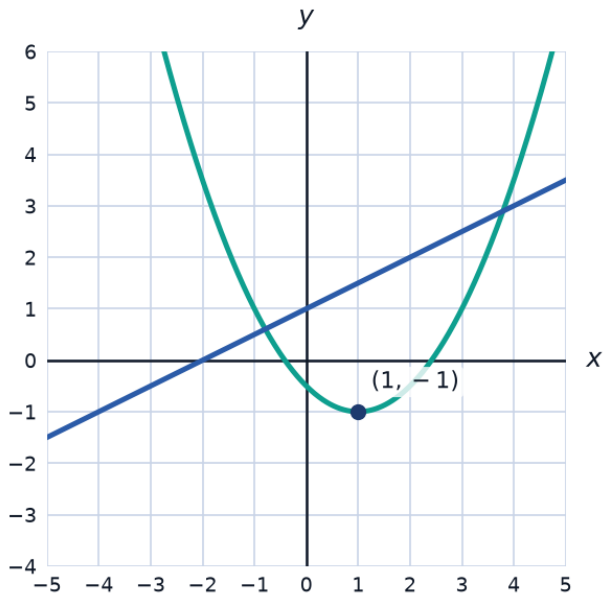
For the arithmetic sequence with $a_1 = 20$ and $d = -3$, find a_{15}.

5 [Foundational] • Target: Arithmetic sequences

A geometric sequence has $a_1 = 3$ and each term is 2 times the previous one. Find the smallest n for which $a_n = 384$.

6 [Foundational] • Target: Arithmetic series (sum formula)

Find S_{12} for the arithmetic sequence with $a_1 = 6$, $d = 4$ (using $a_{12} = 50$ from above).



7 [Foundational] • Target: Arithmetic series (sum formula)

Find S_8 for the arithmetic sequence with $a_1 = 2$, $d = 5$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Geometric sequences

For the geometric sequence with $a_1 = 4$ and common ratio $r = 3$, find a_6.

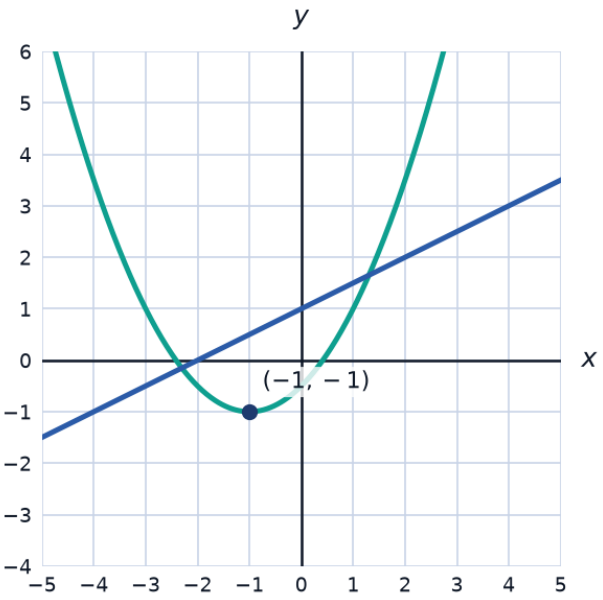
9 [Intermediate] • Target: Geometric sequences

A geometric sequence begins $5, -10, 20, -40, \ldots$. Find r and a_7.

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10 [Intermediate] • Target: Geometric series (finite)

Find S_6 for the geometric sequence with $a_1 = 4$, $r = 3$ (using $a_6 = 972$ from above).



11 [Intermediate] • Target: Geometric series (finite)

Find S_5 for the geometric sequence $5, -10, 20, -40, \ldots$.

12 [Intermediate] • Target: Infinite geometric series

Find the sum of the infinite geometric series $20 + \frac{20}{3} + \frac{20}{9} + \cdots$.

13 [Intermediate] • Target: Infinite geometric series

Find the sum of the infinite geometric series $9 - 3 + 1 - \frac{1}{3} + \cdots$.

14 [Intermediate] • Target: Infinite geometric series

Explain why the infinite geometric series with $a = 5$, $r = 2$ has no finite sum.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

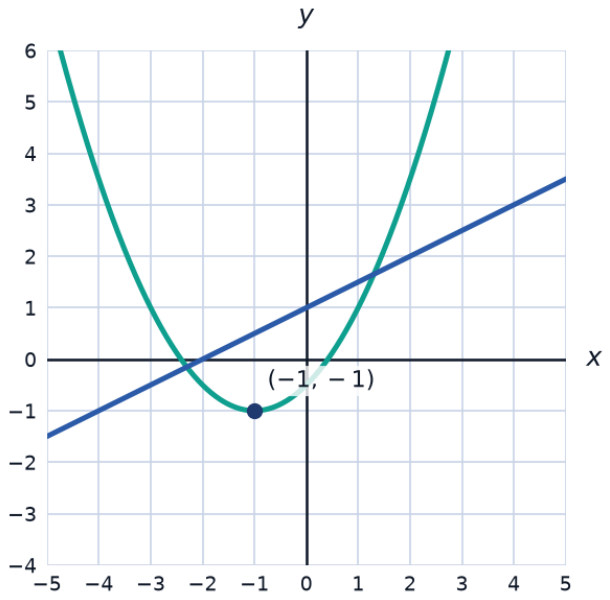
15 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Matrices & Vectors

Find the sum of the infinite geometric series $9 - 3 + 1 - \frac{1}{3} + \cdots$.

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16 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Advanced Functions & Equations

Evaluate the sum of the convergent infinite series: $24 - 12 + 6 - 3 + \cdots$



17 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Evaluate the finite geometric sum: $\sum_{k=1}^{8} 3 \cdot 2^{k-1}$

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Limits

An arithmetic sequence has first term $a_1 = 4$ and common difference $d = 3$. If the sum of the first n terms is 175, find the value of n.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Complex Numbers

In a geometric sequence, $a_1 = 5$ and $a_4 = 40$. Find the common ratio r and the 7th term a_7.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Matrices & Vectors

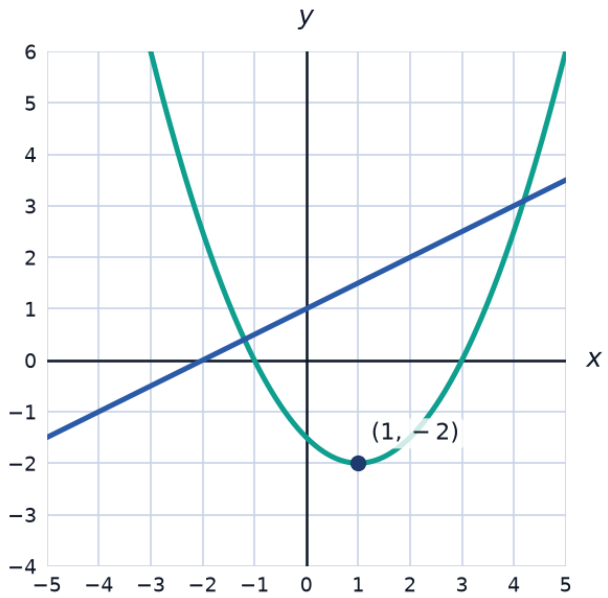
Evaluate the sum: $\sum_{n=1}^{99} \frac{1}{n(n+1)}$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the sum of the infinite geometric series with $a_1 = 7$ and $r = \frac{2}{3}$.



a. 21

b. $\frac{7}{3}$

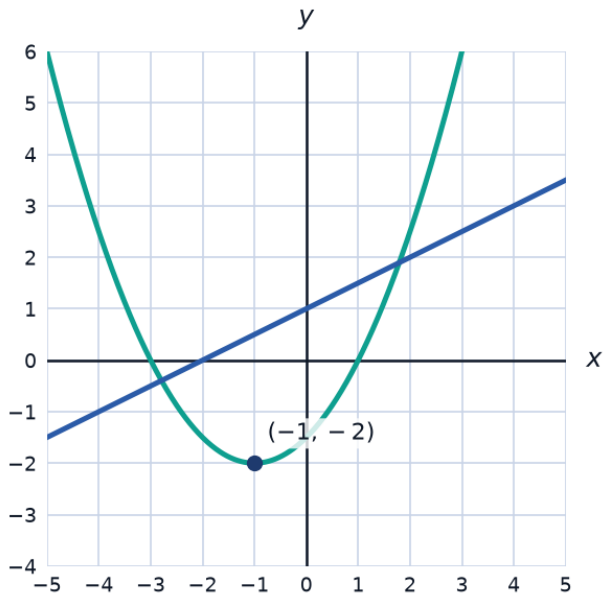
c. $\frac{21}{5}$

d. $\frac{21}{2}$

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is the sum of the telescoping series $\sum_{n=1}^{50} \left(\frac{1}{n} - \frac{1}{n+1}\right)$?



a. $\frac{49}{50}$

b. $\frac{50}{51}$

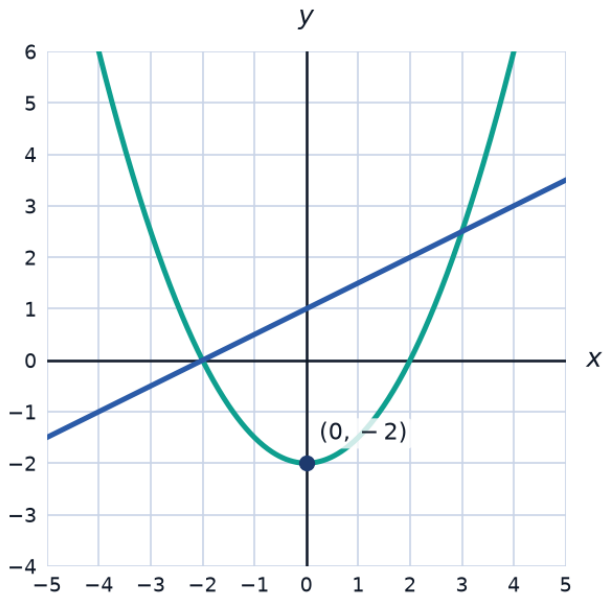
c. 1

d. $\frac{51}{52}$

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23 [Foundational] • Target: Geometric sequences

For the arithmetic sequence with $a_1 = -15$ and common difference $d = 8$, find a_{16}.



a. 113

b. 105

c. 97

d. 107

24 [Foundational] • Target: Geometric series (finite)

For the arithmetic sequence with $a_1 = 2$ and common difference $d = -4$, find a_{21}.

a. -74

b. -78

c. -76

d. -82

25 [Foundational] • Target: Infinite geometric series

For the arithmetic sequence with $a_1 = 18$ and common difference $d = -3$, find a_{18}.

a. -36

b. -33

c. -31

d. -30

26 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the arithmetic sequence with $a_1 = 10$ and common difference $d = 7$, find a_{15}.

a. 108

b. 110

c. 101

d. 115

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27 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the arithmetic sequence with $a_1 = 16$ and common difference $d = 8$, find a_{13}.

a. 114

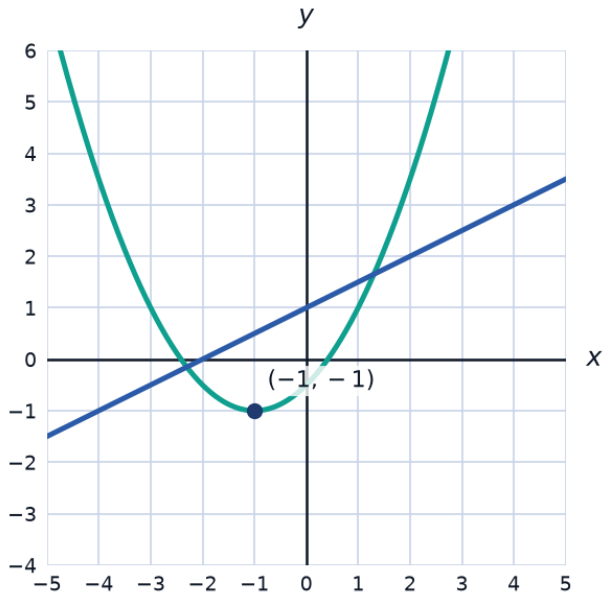
b. 120

c. 104

d. 112

28 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the arithmetic sequence with $a_1 = -5$ and common difference $d = -8$, find a_{21}.



a. −163

b. −157

c. −165

d. −173

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the arithmetic sequence with $a_1 = 3$ and common difference $d = -4$, find a_{21}.

a. −75

b. −81

c. −73

d. −77

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For the geometric sequence with $a_1 = 3$ and common ratio $r = -2$, find a_7.

a. 192

b. -384

c. 190

d. 194

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the geometric sequence with $a_1 = 4$ and common ratio $r = 3$, find a_7.

a. 8748

b. 2913

c. 2916

d. 2919

32 [Intermediate] • Target: Arithmetic sequences

For the geometric sequence with $a_1 = -3$ and common ratio $r = 2$, find a_7.

a. -192

b. -190

c. -194

d. -384

33 [Intermediate] • Target: Arithmetic series (sum formula)

For the geometric sequence with $a_1 = 1$ and common ratio $r = -3$, find a_8.

a. -2184

b. 6561

c. -2190

d. -2187

34 [Intermediate] • Target: Geometric sequences

For the geometric sequence with $a_1 = 1$ and common ratio $r = -2$, find a_4.

a. -8

b. -10

c. 16

d. -6

35 [Intermediate] • Target: Geometric series (finite)

For the geometric sequence with $a_1 = 1$ and common ratio $r = 3$, find a_6.

a. 240

b. 246

c. 243

d. 729

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36 [Intermediate] • Target: Infinite geometric series

For the geometric sequence with $a_1 = 3$ and common ratio $r = -3$, find a_6.

a. -726

b. -732

c. -729

d. 2187

37 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find S_{13} for the arithmetic sequence with $a_1 = -8$, $d = 1$.

a. -27

b. -26

c. -13

d. -4

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Limits

Find S_{12} for the arithmetic sequence with $a_1 = 2$, $d = 9$.

a. 609

b. 630

c. 618

d. 103

39 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Complex Numbers

Find S_{12} for the arithmetic sequence with $a_1 = 7$, $d = -9$.

a. -510

b. -85

c. -501

d. -498

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Matrices & Vectors

Find S_4 for the geometric sequence with $a_1 = 2$, $r = -2$.

a. -10

b. -16

c. -8

d. -12

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41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Advanced Functions & Equations

Find S_6 for the geometric sequence with $a_1 = -3$, $r = 3$.

a. -1095

b. -1092

c. -1089

d. -729

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Find the sum of the infinite geometric series with $a_1 = 7$ and $r = \frac{2}{3}$.

a. 21

b. $\frac{7}{3}$

c. $\frac{21}{5}$

d. $\frac{21}{2}$

43 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 6.01 Limits

Find the sum of the infinite geometric series with $a_1 = 9$ and $r = \frac{2}{3}$.

a. 27

b. 3

c. $\frac{27}{5}$

d. $\frac{27}{2}$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Complex Numbers

Find the sum of the infinite geometric series with $a_1 = 9$ and $r = \frac{2}{3}$.

a. 27

b. 3

c. $\frac{27}{5}$

d. $\frac{27}{2}$

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Matrices & Vectors

Find the sum of all multiples of 4 between 10 and 100 inclusive.

a. $1,188$

b. $1,210$

c. $1,260$

d. $1,320$

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